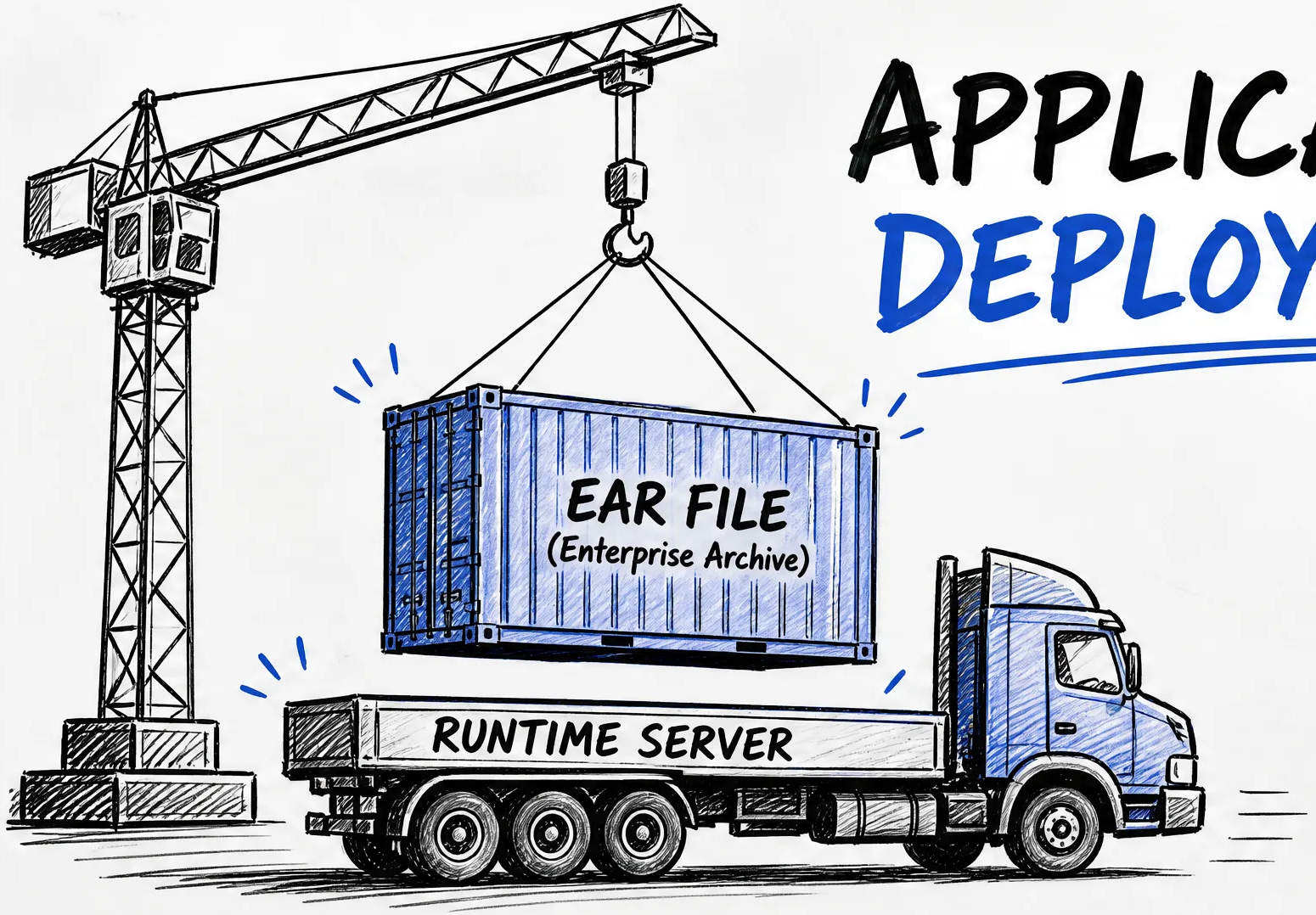
APPLICATION DEPLOYMENT

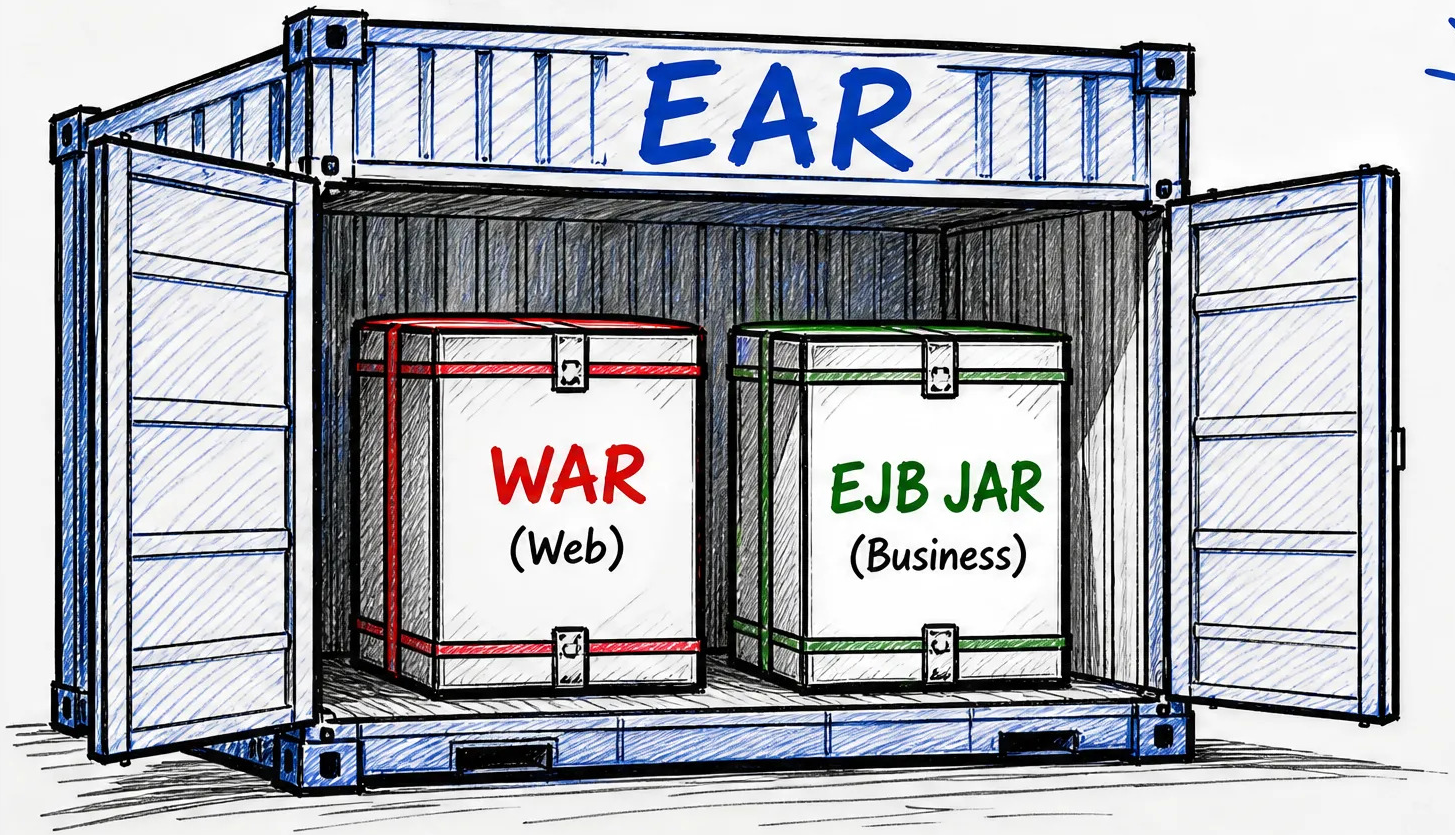


> An archive file becomes a running application only when the platform knows precisely where, how, and with what dependencies to execute it.

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PACKAGE TRUTH:

EAR, WAR, and JAR



ASSEMBLY VS DEPLOYMENT

> The application server deploys the exact artifact it receives; it absolutely does not repair a fundamentally unclear application architecture.

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Preflight the Runtime Contract

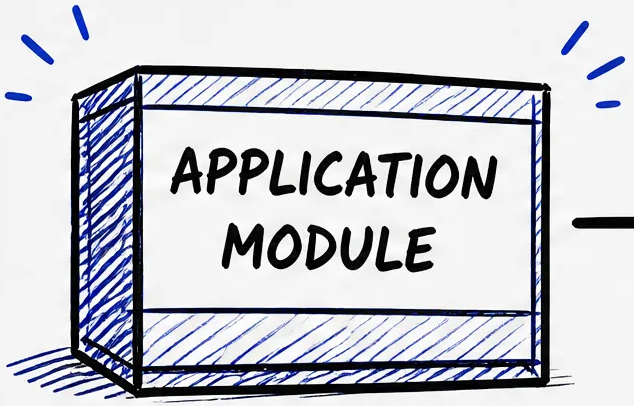
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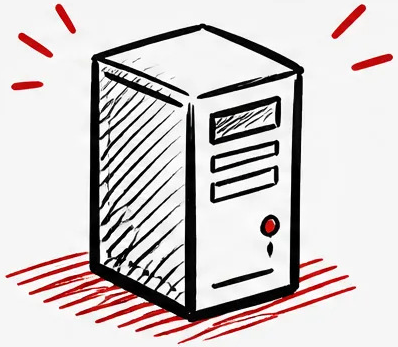
Verify Before You Install

> Proactive preflight work safely turns deployment failure from a massive emergency into a simple checklist item.

CHOOSE THE RIGHT DEPLOYMENT TARGET

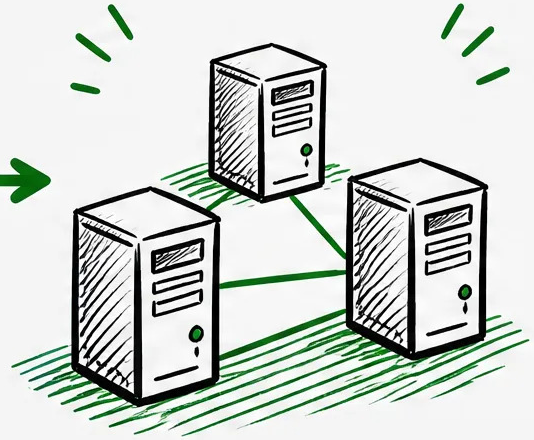


SERVER VS CLUSTER



SERVER

- Isolated
- Single Point of Failure
- Fragile



CLUSTER

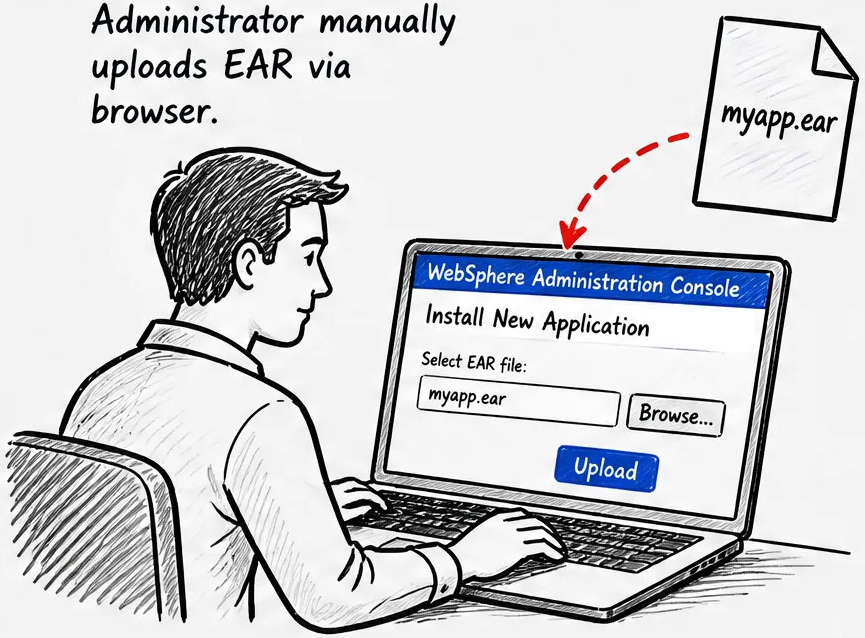
- Interconnected
- High Availability
- Fault Tolerant
- Scalable
- Resilient

> Target mapping is the precise moment an abstract Java module finally becomes a concrete production workload.

Choose a Deployment Channel

MANUAL UPLOAD

Administrator manually uploads EAR via browser.

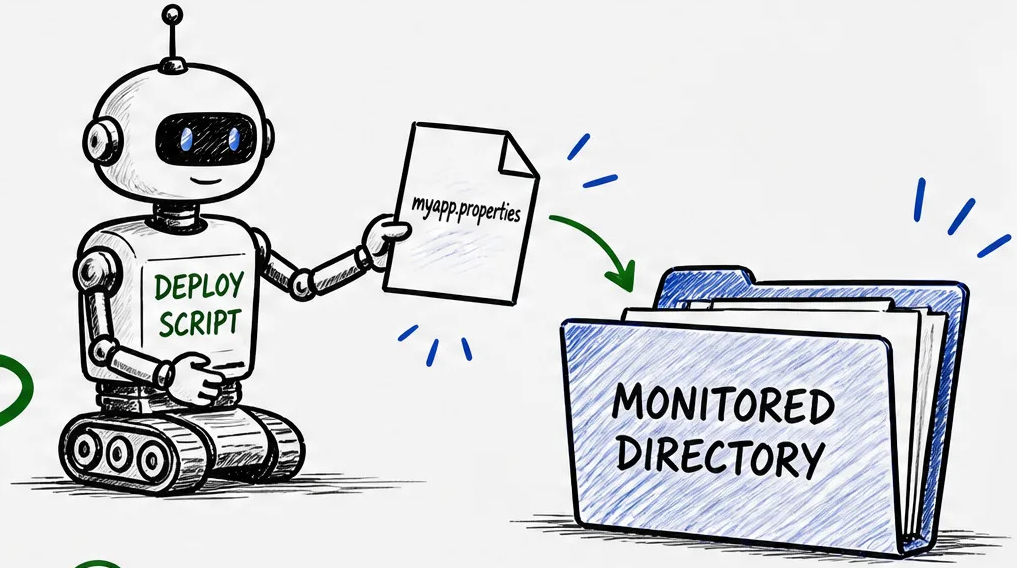


- Manual steps are error-prone and harder to standardize.
- Inconsistent timing and human variability.

MANUAL VS AUTOMATED

AUTOMATED DEPLOYMENT

Automation script places a properties file into a monitored directory.



- Consistent, repeatable, and reliable.
- Reduces manual effort and enables true automation.

> "Artifact location is an essential part of the broader deployment design, because strict repeatability depends completely on knowing exactly what was installed."

The Installation Workflow

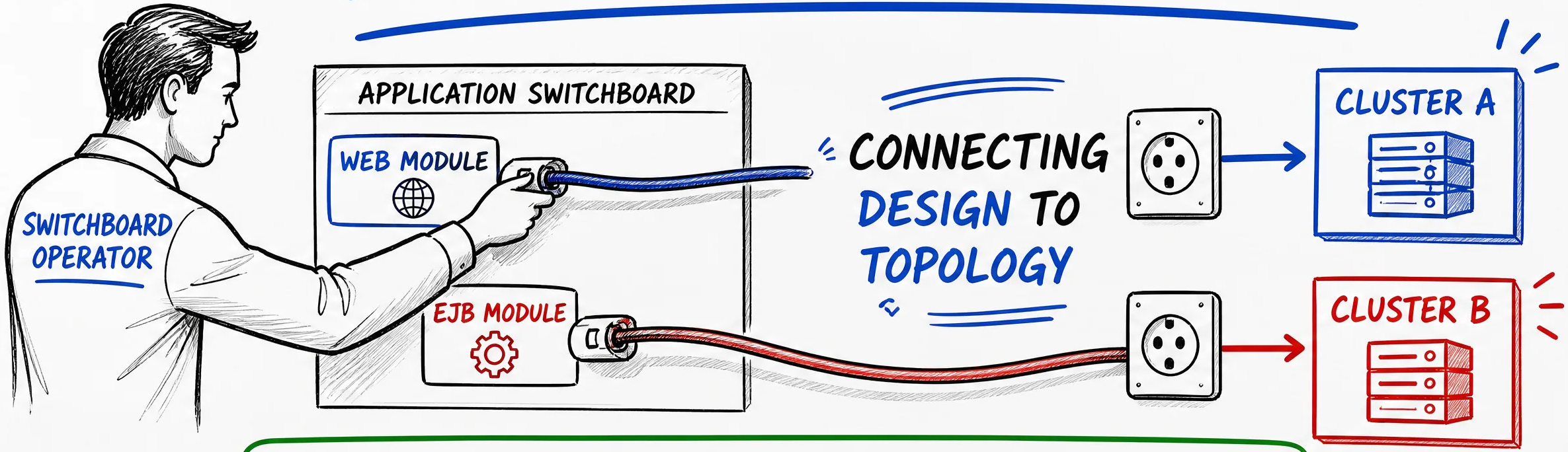


FULLY MANAGED
WEBSPHERE
APPLICATION

Metadata into Runtime Policy

> A deployment definitively has *two phases*:
>
> 1. actively *creating* the configuration
> 2. forcefully *proving* that every target successfully received it.

MAP EVERY MODULE DELIBERATELY



Never lazily treat module mapping as a minor installation checkbox; treat it as a live, highly consequential routing change.

Context Roots and the Public URL

Host:Port



example.com:9080

Where the application is hosted

\+

Context Root



"/trade"

Base path for the web application

\+

Servlet Mapping



"/orders"

How requests are routed to the servlet

=



The Public Front Door

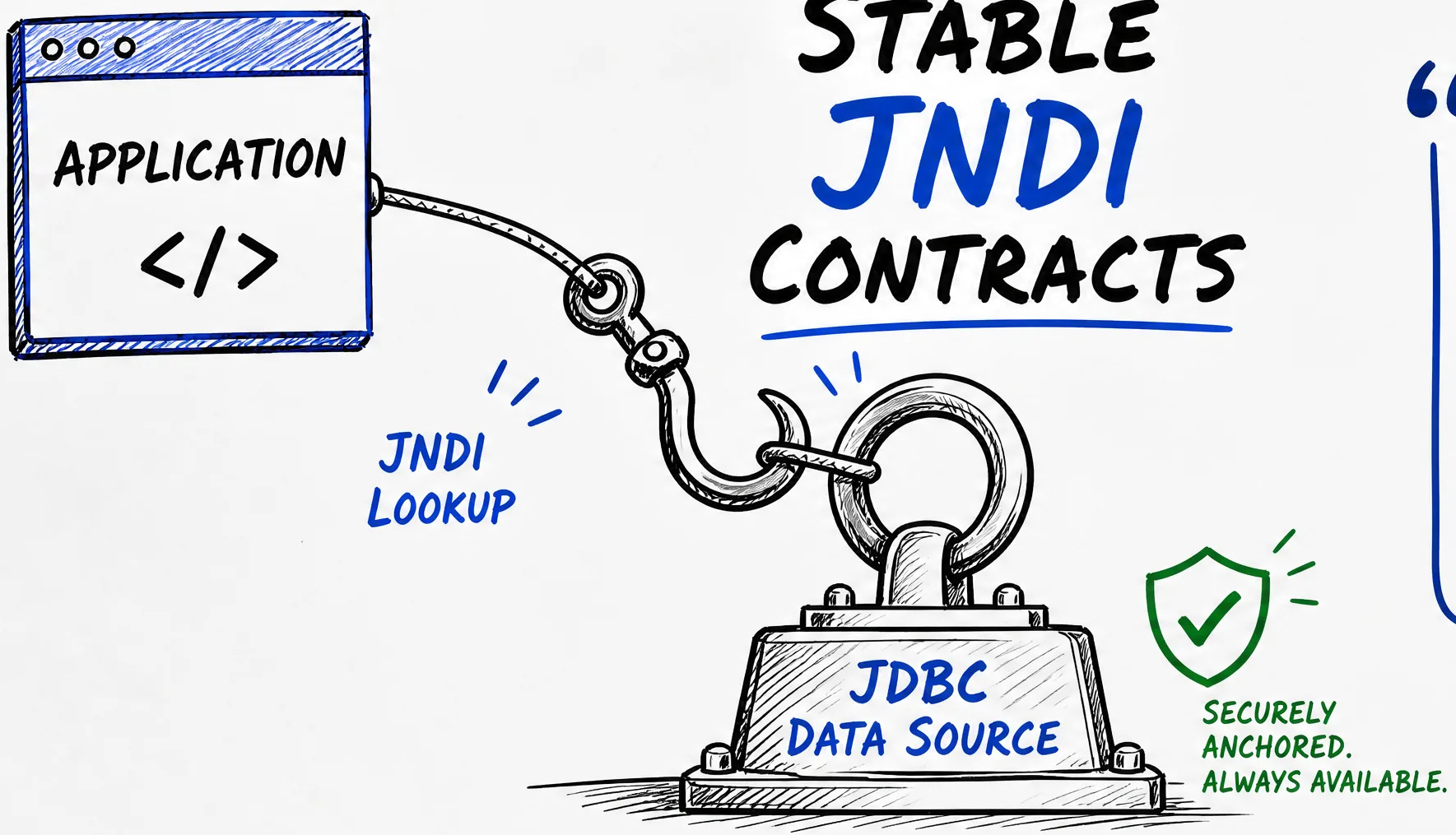
http://example.com:9080/trade/orders



A running web module that cannot be reliably addressed by its intended URL is essentially not deployed from the user's strict perspective.

Resource Readiness: JDBC and JMS

Stable JNDI Contracts



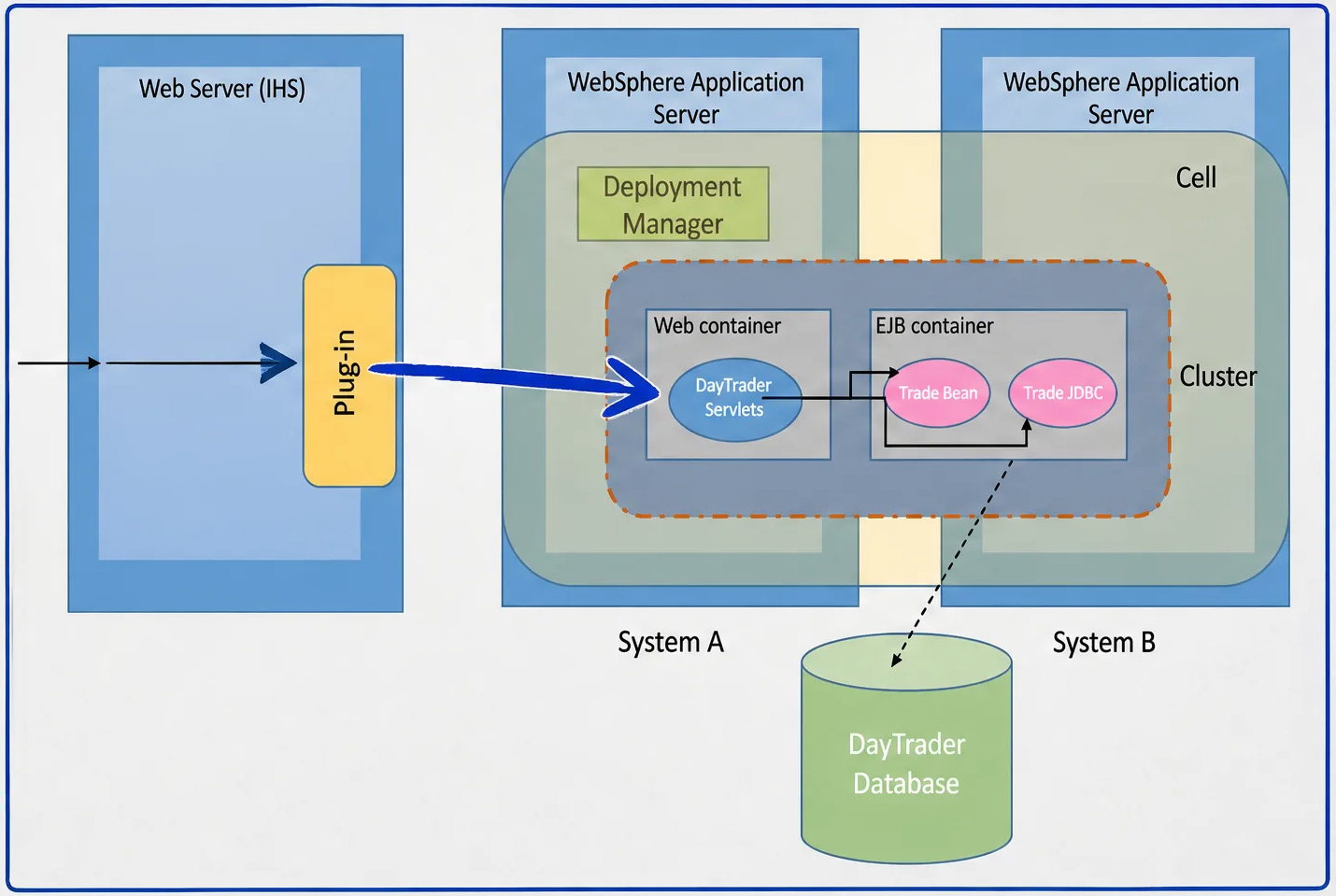
> The absolute fastest deployment diagnosis is definitively proving resources work separately before randomly blaming the new application.

DayTrader as a Deployment Topology

The Complete Route



A successful install only proves the archive arrived; an end-to-end transaction actually proves the system works.

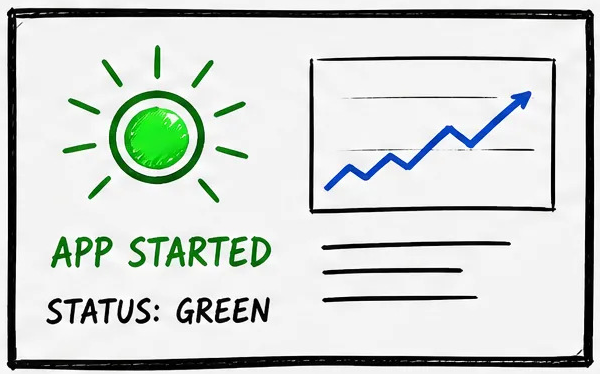


End-to-end confidence = real business value.

Verify, Observe, and Update Safely

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APP DASHBOARD

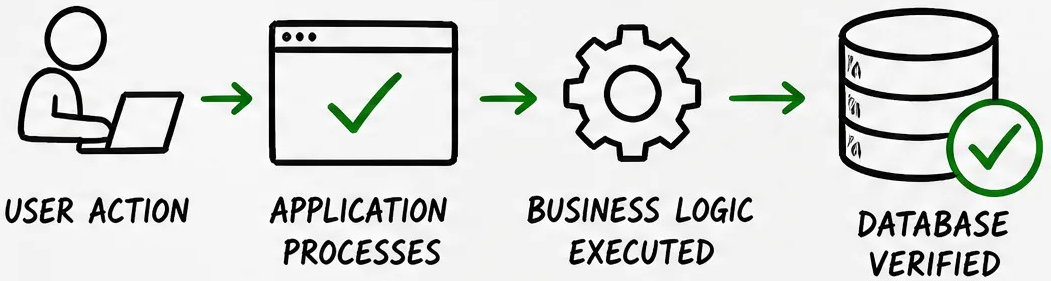


NOT ENOUGH:
A GREEN LIGHT ONLY
MEANS THE APP STARTED.

The Real ACCEPTANCE TEST

THE REAL TEST:

VERIFY END-TO-END SUCCESS



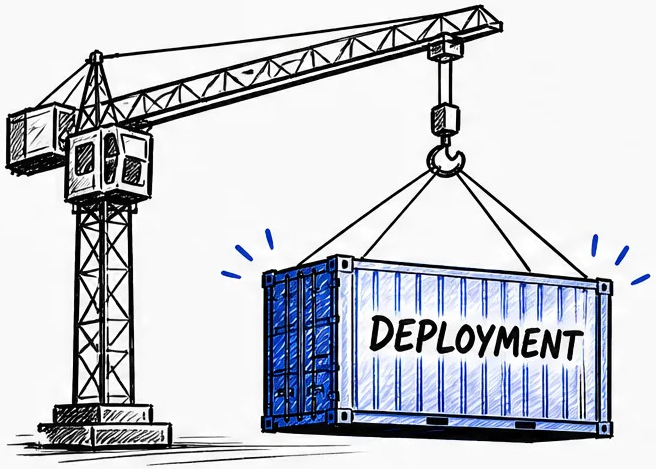
COMPLETE USER TRANSACTION
VERIFIED END-TO-END

> 
> Deployment speed is valuable only when a rollback is significantly faster than customer impact.

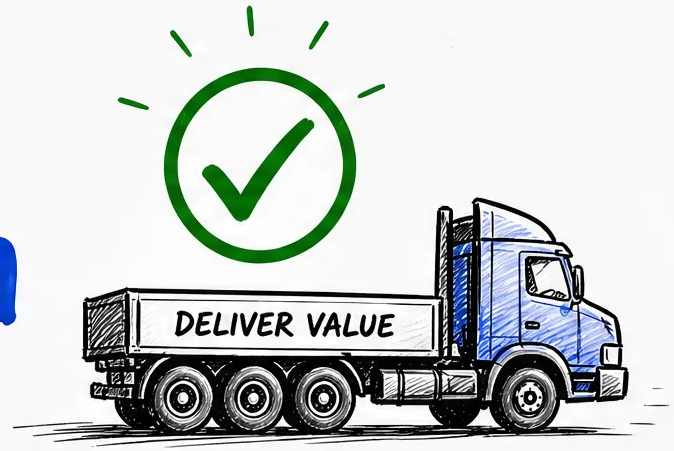
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Module 10 Summary and Lab

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Assembling a Runnable Path



1. PACKAGE
2. PREFLIGHT
3. TARGET
4. CHANNELS
5. WIZARD
6. MAP MODULES
7. CONTEXT ROOT
8. VERIFY

> The administrator's real deliverable is never just an installed file—it is a highly verified, fully supportable application service.